

Smart Metro Ticket Booking System

IDEATION PHASE

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

3. Brainstorming and Idea Prioritization

3.1 Objective

The team held a brainstorming session to surface possible ways to solve the manual, queue-driven metro ticketing problem. The resulting ideas were clustered, assessed, and ranked by how feasible they were on ServiceNow, the impact to the commuter, and the effort needed to build them.

3.2 Raw Brainstormed Ideas

- A digital booking portal on ServiceNow Service Catalog with station and date selection.
- A Flow Designer-driven engine that calculates fare from distance/zone and passenger count.
- QR-coded digital tickets for scanning riders in and out at the gates.
- SMS, email, and push notifications for confirming bookings and delivering tickets.
- Wallet, UPI, and card payment integration for instant checkout.
- Monthly/season pass booking with auto-renewal reminders.
- A live queue/crowd indicator embedded in the booking catalog item.
- Self-service cancellation and refund workflow for tickets.
- A chatbot / Virtual Agent to help with bookings and common questions.
- A staff-facing dashboard for tracking bookings, revenue, and station traffic.
- A voice-assisted accessibility mode for elderly and differently-abled riders.
- Group booking for families or corporate travel.
- IoT/gate-hardware integration for automatic QR scanning and validation.
- An analytics/reporting module to forecast peak hours and tune fares.

3.3 Idea Grouping (Affinity Mapping)

Theme	Grouped Ideas
Core Booking Experience	Service Catalog booking form, station/date/passenger selection
Communication & Delivery	Notifications (SMS/Email/Push), booking confirmation
Payments	UPI/Card/Wallet integration, season pass renewal
Operations & Admin	Admin dashboard, analytics/reporting, gate hardware integration
Support & Accessibility	Virtual Agent/chatbot, voice-assisted accessibility mode, cancellation

3.4 Prioritization Matrix (Impact vs. Effort)

Idea	Customer Impact	Implementation Effort	Priority
Service Catalog based ticket booking form	High	Low	P0 - Must Have
Automated fare calculation (Flow Designer)	High	Medium	P0 - Must Have
QR-code digital ticket generation	High	Low	P0 - Must Have

Idea	Customer Impact	Implementation Effort	Priority
Booking confirmation notifications (SMS/Email)	High	Low	P0 - Must Have
Payment gateway integration	High	Medium	P1 - Should Have
Ticket cancellation/refund workflow	Medium	Medium	P1 - Should Have
Admin dashboard for staff/operations	Medium	Medium	P1 - Should Have
Season pass booking & renewal	Medium	Medium	P2 - Could Have
Virtual Agent/chatbot support	Medium	High	P2 - Could Have
Gate hardware/IoT scanner integration	Medium	High	P2 - Could Have
Accessibility voice-assistance mode	Medium	High	P2 - Could Have
Analytics & peak-hour prediction	Low	High	P3 - Future Scope

3.5 Finalized Ideas for Phase 1 (MVP)

Going by the prioritization matrix, the team locked in the following as the Minimum Viable Product (MVP) scope, all buildable using core ServiceNow components:

- A Service Catalog item to pick source station, destination station, travel date, and passenger count.
- A Flow Designer workflow to work out the fare automatically from station distance/zone and passenger count.
- Automatic issuing of a unique digital ticket with an embedded QR code once booking succeeds.
- A notification step to send the digital ticket and confirmation to the commuter right away.
- Basic tracking of each user's booking history/status via the Service Catalog request record.